

Vivado HLS 2019.1 - Troubleshooting

Unknow File

Description

While running C Synthesis Vivado returned an error saying it was not able to find the file `matrix_mult.h`. Turns out that the test bench files created by Vivado were stored in the `AppData/Roaming/Xilinx/Vivado` folder not in the project folder.

Solution

This issue was resolved by moving test bench files to the project folder and including those to the project overwriting files.

The screenshot shows the Vivado HLS 2019.1 interface for a project named 'matrix_multiplication'. The Explorer pane on the left shows the project structure, including 'matrix_mult.cpp', 'matrix_mult.h', and 'matrix_mult_tb.cpp'. The 'Test Bench' folder is highlighted, and a context menu is open with options like 'Add Files...', 'New File...', and 'Add Folder...'. The main pane displays 'Performance Estimates' for 'matrix_mult.cpp'. It includes a 'Timing (ns)' summary table, a 'Latency (clock cycles)' summary table, and a 'Detail' section for the 'Loop'.

Clock	Target	Estimated	Uncertainty
ap_clk	10.00	7.256	1.25

Latency	Interval			
min	max	min	max	Type
6148	6148	6148	6148	none

Loop Name	min	max	Iteration Latency	achieved	target	Trip Count	Pipelined
- loop_stream_read_row_A	80	80	10	-	-	8	no
+ loop_stream_read_col_A	8	8	1	-	-	8	no
- loop_stream_read_row_B	80	80	10	-	-	8	no
+ loop_stream_read_col_B	8	8	1	-	-	8	no
- loop_mult_row	5776	5776	722	-	-	8	no
+ loop_mult_col	720	720	90	-	-	8	no
++ loop_mult_k	88	88	11	-	-	8	no
- loop_stream_write_row	208	208	26	-	-	8	no
+ loop_stream_write_col	24	24	3	-	-	8	no

The 'Utilization Estimates' section shows a summary table with columns for Name, BRAM, 18K, DSP4RE, FF, LUT, and URAM. The Console pane at the bottom shows 1 DRC-Infos, 0 DRC-Warnings, and 0 DRC-Errors. A message is displayed: '[SCHED 204-61] Option 'relax_ii_for_timing' is enabled, will increase II to preserve clock frequency constraints.'

Export IP Invalid Argument

Description

During execution of Export IP it may return an error saying `Revision Number Overflow` which is caused by the machine datetime being beyond what is expected.

Solution

An external correction patch is required to fix this issue as explained in [Export IP Invalid Argument / Revision Number Overflow Issue \(Y2K22\) \(amd.com\)](#).

Software Datetime

Description

It happens because there is an error in the datetime of the software that simply breaks if you the computer is in the present.

Solution

Modify computer datetime to before launch date of the software, around January of 2019.